

# Metric-Induced Stability Beyond Spectral Criteria in Gated Higher-Order Dynamical Systems

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## Abstract

We study Gated Higher-Order Dynamical (GHOD) systems of the form  $\dot{x} = -Ax - g(t)\nabla \Phi(x)$  through a single central object: the effective quadratic operator  $\bar{G}(x,t) = A + g(t)H(x)$ , where  $H(x) = \nabla^2 \Phi(x)$ . The fundamental observation is an exact decomposition  $\dot{x} = -\bar{G}(x,t)x - g(t)r(x)$ , where  $r(x) = \nabla \Phi(x) - H(x)x$  captures all nonlinear effects beyond the quadratic approximation. Global stability emerges from the competition between the quadratic operator  $\bar{G}$ , which governs local behaviour, and the nonlinear remainder  $r(x)$ , which enforces global dissipation.

We establish four main theorems. Theorem 1 (Metric-Nonlinear Stabilisation): global boundedness and compact attractor existence hold under a superquadratic restoring condition on  $r(x)$ , even when  $\bar{G}(x,t)$  has negative eigenvalues — local linear instability does not preclude global boundedness. Theorem HG (Metric-Hypocoercivity): under stochastic regularisation with time-varying  $g(t)$ , exponential entropy decay persists under the unified metric condition  $\|g - \bar{g}\|_{\infty} \cdot L < 2\epsilon \sqrt{\mu_{LS}}$ , proved without any antisymmetry assumption on the non-gradient perturbation. Theorem H-LSI (Log-Sobolev upgrade): rate  $\lambda_{HG} = 2\epsilon \mu_{LS} - C_{\bar{G}}^2/(2\epsilon)$ , with clean symmetric dependence on  $\epsilon$ . Theorem S (Spectral Resolution): the transport-corrected operator  $J_{\text{eff}} = J + A_{\bar{G}}$  satisfies  $\lambda_{GHOD} < 0$  under the same metric condition, explaining why local positive eigenvalues of  $J = -\bar{G}$  do not produce global instability.

The Bakry-Emery curvature of the system equals  $\bar{G}$  in the gradient case, connecting the stability framework to information-geometric convergence theory. Four conjectures propose extensions to holonomy theory (holonomy group compact iff Governor Inequality holds), Wasserstein geometry (free energy displacement convex iff  $\bar{G}$  positive definite), BCM synaptic plasticity (BCM implements natural gradient descent on the GHOD Fisher manifold), and pre-ictal EEG dynamics (condition number  $\kappa(t) \sim (T^*-t)^{-1}$  before seizure onset; the exponent -1 follows by direct algebra from the OU-alpha model — the conjecture is whether this model describes real pre-ictal dynamics, an empirical question under active testing). Four errors from a previous draft of the hypocoercivity argument are corrected and documented in Section 8. All theorems are proved under explicit assumptions. Conjectures are labelled as such. Geometric constructions are labelled as interpretative.

**Scope statement:** *This is an exploratory theoretical note. All mathematical results are internally derived and reproduced in simulation. No external peer review has been conducted. Assumption (A5) must be verified case-by-case for specific systems; an explicit verification for the cubic p-spin interaction is provided in Section 2.3. Empirical validation of Conjecture 4 against the CHB-MIT EEG database is in progress; current status is documented in Section 11.4. The work is made public to expose the framework to the research community and invite scrutiny, not to assert a completed theory.*

## 1. Introduction

Stability of nonlinear dynamical systems is often assessed through local spectral information: if the Jacobian DF of a vector field has eigenvalues with positive real part at a point, local instability is expected. This criterion is sufficient but not necessary. Nonlinear systems may maintain global stability despite locally unstable linearisations when higher-order dissipative structure dominates at large scales. This is a known phenomenon in nonlinear dynamics — Lyapunov functions routinely override local spectral predictions — and the present paper gives the explicit mechanism for the GHOD class.

Gated Higher-Order Dynamical systems [1] document this phenomenon concretely: Regime B of [1] shows globally stable behaviour with spectral ratios  $10^4$  times the classical Lyapunov bound. The explanation requires going beyond linearisation.

Independent convergent evidence comes from Goto, Lopez Rios, Scholz, and Vaikuntanathan [16] (arXiv:2512.13859, December 2025), who introduce a neuromodulation-inspired gated associative memory where neuropeptide-like signals are modelled by a self-adaptive activity-dependent gating mechanism. Using dynamical mean-field theory and many-body simulations at scale, they show that such gating fundamentally reorganises the attractor structure: the network bypasses the classical spin-glass transition and maintains robust high-overlap retrieval far beyond the standard Hopfield critical capacity, without shrinking basins of attraction. This is convergent evidence for the core GHOD insight that gating can decouple capacity from stability (developed concurrently and independently: Goto et al. appeared December 2025; the present framework was developed January-February 2026 building on the companion CONN paper [1] from December 2025). The two approaches are complementary: Goto et al. provide large-N DMFT rigour, biological motivation, and capacity measurements; the present work provides explicit Lyapunov stability guarantees (Theorem 1), the Metric-Nonlinear Stabilisation mechanism (the  $G\text{-}r(x)$  decomposition), and quantitative convergence rates via hypocoercivity theory. The gating mechanisms differ: Goto et al. use multiplicative gating modulating effective neuronal time-constants; GHOD uses additive decaying gating of higher-order interaction terms. A rigorous comparison of the two stability mechanisms remains an open direction.

The organising object is the effective quadratic operator:

$$G(x,t) := \nabla_x^2 E(x,t) = A + g(t) * H(x), \quad H(x) = \nabla_x^2 \Phi(x) \text{ (effective quadratic operator)}$$

Note:  $G(x,t)$  is called an effective quadratic operator, not a metric, throughout. It is a Riemannian metric only in the subcritical regime  $\beta < 1$  when  $G$  is positive definite. In the indefinite regime the geometric word "metric" is avoided. The key structural result is the exact decomposition:

$$\dot{x} = -G(x,t) * x - g(t) * r(x) \quad \text{where } r(x) = \nabla \Phi(x) - H(x) * x \quad \text{(D) exact}$$

This separates two mechanisms:  $-G(x,t)*x$  governs local quadratic behaviour (dissipative when  $G$  positive definite, locally expanding when  $G$  indefinite);  $-g(t)*r(x)$  is the pure nonlinear correction, invisible to linearised analysis. Stability arises from the superquadratic domination of  $r(x)$  at large  $\|x\|$ , regardless of the sign of  $G$ .

Relation to concurrent work: Goto et al. (2025, arXiv:2512.13859) independently propose a neuromodulation-inspired gated associative memory where a self-adaptive activity-dependent gate bypasses the classical spin-glass transition and maintains high-overlap retrieval far beyond the Hopfield capacity limit, using many-body simulations and dynamical mean-field theory at  $N=1000$ . The core insight — that gating reorganises the attractor structure and permits operation beyond classical stability limits — is convergent with the GHOD framework developed independently here. The two approaches are complementary: Goto et al. provide large-scale empirical and DMFT validation; GHOD provides explicit Lyapunov stability guarantees (Theorem 1) and the exact Lyapunov-level mechanism via the  $G(x,t)*x + g(t)*r(x)$  decomposition. The gating mechanisms differ: Goto et al. use multiplicative neuromodulatory gating on effective time-constants; GHOD uses a decaying temporal gate on higher-order interaction terms with explicit return guarantees.

Paper structure: Section 2 (setup and decomposition), Section 3 (Theorem 1, complete proof), Section 4 (global attractor), Section 5 (invariant measures), Section 6 (entropy convergence), Section 7 (Bakry-Emery), Section 8 (Theorem HG and H-LSI, hypocoercivity, four corrections), Section 9 (Theorem S, spectral resolution), Section 10 (gated connection, interpretative), Section 11 (four conjectures), Section 12 (limitations), Section 13 (conclusion), Appendices A-C (complete proofs).

## 1.1 Reader's Guide

Given the conceptual density of this work, readers from different backgrounds are directed to the sections most relevant to their interests:

- Dynamical systems / nonlinear stability: Sections 2-4 (setup, Theorem 1, attractor) and Section 9 (spectral resolution). These sections require only standard ODE theory and Lyapunov methods. The core result — global stability despite locally positive eigenvalues — is self-contained in Sections 2-3.
- Stochastic analysis / information geometry: Sections 5-7 (invariant measures, entropy, Bakry-Emery) and Section 8 (hypocoercivity). These sections use Fokker-Planck equations, Fisher information, and Log-Sobolev inequalities. Section 8 is self-contained with complete proofs in Appendices A-B.
- Differential geometry / connections: Section 10 (gated connection, curvature). This section is explicitly interpretative and can be read independently. It does not contain theorems whose conclusions appear elsewhere.
- Neuroscience / clinical applications: Section 11 (conjectures), especially Conjectures 3-4 (BCM and pre-ictal EEG). The mathematical status of each conjecture is stated explicitly. Section 11.4 contains the analytical proof that the seizure-precursor exponent equals 1 exactly, and the current empirical validation protocol.
- All readers: the Limitations section (Section 12) should be read before drawing conclusions. It lists five specific technical gaps honestly.

## 2. Setup, Notation, and Exact Decomposition

### 2.1 Standing assumptions

Throughout the paper:

- (A1)  $A$  in  $\mathbb{R}^{(n \times n)}$ , symmetric positive definite,  $\lambda_{\min}(A) > 0$ .
- (A2)  $g$  in  $L^{\infty}([0, \infty))$  with  $|g(t)| \leq g_{\max} < \infty$ .
- (A3)  $\Phi$  in  $C^3(\mathbb{R}^n)$ ,  $\|H(x)\|_{\text{op}} = \|\nabla^2 \Phi(x)\|_{\text{op}} \leq L$  for all  $x$ .
- (A4)  $\nabla \Phi$  is  $C^2$ -Lipschitz with constant  $L$  (from A3).

Under (A1)-(A4),  $F(x, t) = -Ax - g(t)\nabla \Phi(x)$  is globally Lipschitz in  $x$ , uniformly in  $t$ . Unique global solutions exist for all initial data by Caratheodory.

### 2.2 Effective quadratic operator

Define:

$$E(x, t) = (1/2) x^T A x + g(t) \Phi(x)$$

$$G(x, t) := \nabla_x^2 E(x, t) = A + g(t) H(x) \text{ (effective quadratic operator)}$$

Eigenvalues:  $\lambda_i(G) = \lambda_i(A) + g(t)\lambda_i(H(x))$ . Governor Inequality:  $\beta := g_{\max}L / \lambda_{\min}(A) < 1$ . Under this condition  $G$  is uniformly positive definite with  $\lambda_{\min}(G) \geq \Delta := \lambda_{\min}(A) - g_{\max}L > 0$ . When  $\beta \geq 1$ ,  $G$  may have indefinite signature.

**Remark.**  $G(x,t)$  is an effective quadratic operator. The term metric applies only when  $G$  is positive definite ( $\beta < 1$ ). In the indefinite case ( $\beta > 1$ ),  $G$  has pseudo-Riemannian signature and geometric conclusions requiring positive-definiteness must not be imported.

## 2.3 Exact dynamical decomposition

Define:

$$r(x) := \text{nabla} \Phi(x) - H(x) * x \text{ (nonlinear remainder)}$$

For  $\Phi$  in  $C^3$  with  $\text{nabla} \Phi(0) = 0$ :  $r(x) = O(\|x\|^2)$ . Generally:  $\|r(x)\| \leq \|\text{nabla} \Phi(0)\| + C_r \|x\|^2$ . Then (GHOD) is exactly:

$$\dot{x} = -G(x,t) * x - g(t) * r(x) \quad (D)$$

This is algebraically exact. The term  $-G(x,t)*x$  is the quadratic component. The term  $-g(t)*r(x)$  is the pure higher-order nonlinear correction, invisible to linearised analysis.

## 2.4 Explicit verification of assumption (A5) for the cubic p-spin interaction

The nonlinear restoring condition (A5) — the core assumption of Theorem 1 — must be verified for specific systems. We give the explicit computation for the primary GHOD use case: the cubic p-spin interaction with  $p = 3$ .

Let  $\Phi(x) = (1/6) * T_{\{ijk\}} x_i x_j x_k$  with  $T$  fully symmetric. Then:

$$\begin{aligned} \text{nabla}_i \Phi(x) &= (1/2) T_{\{ijk\}} x_j x_k && \text{(gradient of cubic)} \\ H_{\{ij\}}(x) &= \text{nabla}^2_{\{ij\}} \Phi(x) = T_{\{ijk\}} x_k && \text{(Hessian, linear in } x) \end{aligned}$$

Compute the nonlinear remainder  $r(x) = \text{nabla} \Phi(x) - H(x)*x$ :

$$r_i(x) = (1/2) T_{\{ijk\}} x_j x_k - T_{\{ijk\}} x_k x_j = -(1/2) T_{\{ijk\}} x_j x_k$$

Therefore  $r(x) = -(1/2) \text{nabla} \Phi(x)$ . The inner product  $\langle x, r(x) \rangle$  is:

$$\begin{aligned} \langle x, r(x) \rangle &= -(1/2) \langle x, \text{nabla} \Phi(x) \rangle = -(1/2) * (3/6) \sum_{\{ijk\}} T_{\{ijk\}} x_i x_j x_k \\ &= -(1/4) T_{\{ijk\}} x_i x_j x_k && \text{(cubic form in } x) \end{aligned}$$

Assumption (A5) requires  $\langle x, r(x) \rangle \geq \beta_0 * \|x\|^3$  for large  $\|x\|$  (here  $\delta = 1$ ). This holds when the cubic form  $T_{\{ijk\}} x_i x_j x_k$  is negative for large  $\|x\|$  in the radial direction. The sign depends on the tensor  $T$ :

- If  $T$  is chosen so that  $T_{\{ijk\}} x_i x_j x_k > 0$  for large  $\|x\|$  (positive cubic), then  $\langle x, r(x) \rangle = -(1/4) * T_{\{ijk\}} x_i x_j x_k < 0$ , and (A5) holds with  $\beta_0 > 0$  and  $\delta = 1$ .
- If  $T$  is a generic symmetric tensor — meaning outside the measure-zero set where the cubic form  $T_{\{ijk\}} x_i x_j x_k$  is identically zero in the radial direction — then (A5) holds. For GOE random tensors ( $T_{\{ijk\}}$  i.i.d. standard normal, symmetrised), the cubic form has nonzero radial component almost surely: the sign is fixed once  $T$  is drawn, and the condition fails only on a set of Lebesgue measure zero in the space of symmetric tensors. The probabilistic language is a convenience; the precise statement is that (A5) holds generically, i.e. for all tensors outside a measure-zero exceptional set.
- General condition: (A5) holds for the cubic p-spin interaction whenever the cubic form satisfies  $T_{\{ijk\}} n_i n_j n_k > 0$  where  $n = x/\|x\|$  is the radial direction. This is a non-degeneracy condition on  $T$  satisfied by all tensors outside a measure-zero set.

**Remark.** This computation also shows that for p-spin with  $p = 3$ , the remainder  $r(x) = -(1/2) * \text{nabla} \Phi(x)$  scales as  $\|x\|^2$  for large  $\|x\|$  — confirming that (A5) holds with  $\delta = 1$  (one power above quadratic). The Lyapunov proof in Theorem 1 therefore gives  $V_{\dot{}} \leq C_1 * \|x\|^2 - C_2 * \|x\|^3$ , which is negative for  $\|x\| > C_1/C_2$ , establishing the compact absorbing set with radius  $K_* \sim C_1/C_2$ .

## 2.5 Structural conditions for Assumption (A5)

The p-spin verification answers "does (A5) hold for this specific system?" A practitioner also needs to know: "how do I check (A5) for my system?" The following proposition gives three sufficient structural conditions, each checkable from the definition of  $\Phi$  without explicit computation of  $r(x)$ .

### Proposition SC — Sufficient Conditions for (A5).

Let  $\Phi$  in  $C^3(R^n)$ . Each of the following conditions independently implies (A5):

(SC1) Radial superquadratic growth with positive sign:

There exist  $\beta_0 > 0$ ,  $\delta > 0$ ,  $R > 0$  such that

$\langle x, \nabla \Phi(x) \rangle \geq \beta_1 \|x\|^{2+\delta}$  for all  $\|x\| \geq R$ .

Then  $r(x) = \nabla \Phi - H(x)x$  satisfies (A5) with the same  $\delta$ ,

because  $\langle x, r(x) \rangle = \langle x, \nabla \Phi(x) \rangle - x^T H(x) x \geq \beta_1 \|x\|^{2+\delta} - L \|x\|^2$ , which is positive for large  $\|x\|$  when  $\delta > 0$ .

(SC2) Radial monotonicity of the gradient:

$\Phi$  is radially convex in the sense that  $d/dr [\langle x, \nabla \Phi(x) \rangle] > 0$

for  $\|x\| = r$  large, where  $r = \|x\|$ . Then the cubic (or higher) growth of  $\langle x, \nabla \Phi \rangle$  dominates the quadratic  $x^T H x$ , and (A5) holds with  $\delta$  equal to the degree of superquadratic growth minus 2.

(SC3) Homogeneous potential of degree  $p \geq 3$ :

If  $\Phi$  is homogeneous of degree  $p \geq 3$  ( $\Phi(\lambda x) = \lambda^p \Phi(x)$ ),

then by Euler's theorem:  $\langle x, \nabla \Phi(x) \rangle = p \Phi(x)$ .

And  $x^T H(x) x = p(p-1) \Phi(x)$  (from differentiating Euler).

Therefore:  $\langle x, r(x) \rangle = \langle x, \nabla \Phi(x) \rangle - x^T H(x) x$

$$= p \Phi(x) - p(p-1) \Phi(x)$$

$$= p(2-p) \Phi(x).$$

For  $p \geq 3$  and  $\Phi(x) > 0$  for large  $\|x\|$ :  $\langle x, r(x) \rangle = p(2-p) \Phi < 0$

requires  $\Phi < 0$ . If instead  $\Phi < 0$  for large  $\|x\|$  (confining potential),

then  $\langle x, r(x) \rangle = p(2-p) \Phi(x) \geq \beta_0 \|x\|^p$  with  $\beta_0 > 0$ ,

so (A5) holds with  $\delta = p - 2 \geq 1$ .

Practical test: for any specific  $\Phi$ , compute  $\langle x, \nabla \Phi(x) \rangle$  along the radial direction  $x = r e_1$  and check whether it grows faster than  $r^2$ .

If yes, (A5) holds with  $\delta = (\text{degree of growth}) - 2 > 0$ .

These three conditions cover the main cases arising in practice. (SC1) is the most general; it directly verifies (A5) via the gradient. (SC2) provides a geometric criterion based on the shape of the potential. (SC3) is the sharpest and most elegant: for any homogeneous potential of degree  $p \geq 3$  that is confining (negative for large  $\|x\|$ ), (A5) holds automatically with  $\delta = p - 2$ . The cubic p-spin verification in Section 2.4 is a special case of (SC3) with  $p = 3$ .

**Remark.** When none of (SC1)-(SC3) applies, (A5) must be verified directly. The direct check is: compute  $r(x) = \nabla \Phi(x) - H(x)x$  explicitly, then verify  $\langle x, r(x) \rangle$  grows faster than  $\|x\|^2$  along the relevant directions. If (A5) fails entirely — e.g. for quadratic  $\Phi$  where  $r(x) = 0$  identically — Theorem 1 reduces to the classical subcritical case (Corollary to Theorem 1) where stability follows from the Governor Inequality alone.

## 2.6 Explicit Confinement Radius and the Three-Regime Classification

Theorem 1 establishes the existence of a compact absorbing set  $K_*$  but does not compute its radius explicitly. The following lemma fills this gap, giving  $R^*$  as a closed-form expression in the system parameters and revealing why three distinct stability regimes emerge naturally from the balance between dissipation and nonlinear restoring forces.

### Lemma CR — Confinement Radius and Regime Classification.

*Notation: throughout this lemma,  $\Delta := \lambda_{\min}(A) - g_{\max}L$  denotes the Governor Inequality margin (spectral dissipation margin), distinct from the superquadratic growth exponent  $\delta$  in Assumption (A5).  $C_2 := g_{\max} \cdot \beta_0$  is the nonlinear restoring amplitude. Under (A1)-(A5) with  $\Delta > 0$  (Regime B):*

$$\begin{aligned} R^* &= (\Delta / C_2)^{1/\delta} \\ &= (\lambda_{\min}(A) - g_{\max}L) / (g_{\max} \cdot \beta_0)^{1/\delta} \end{aligned}$$

*where  $\delta$  is the growth exponent from (A5) and  $\Delta$  is the spectral margin.*

*For the cubic p-spin case ( $\delta = 1$ ):  $R^* = \Delta / C_2 = (\lambda_{\min}(A) - g_{\max}L) / (g_{\max} \cdot \beta_0)$ .*

*Properties (Regime B only,  $0 < \Delta$  small):*

- (i)  $R^*$  is computable from system parameters alone, without simulation.
- (ii) Every trajectory eventually enters  $\{\|x\| \leq R^*\}$  and remains there.
- (iii)  $R^* \rightarrow 0$  as  $\Delta \rightarrow \inf$  (Regime A limit: absorbing set collapses to origin).
- (iv)  $R^*$  is undefined when  $\Delta \leq 0$  (Regime C: no finite confinement radius).

*The three stability regimes arise from the size of  $\Delta$ :*

*Regime A ( $\Delta \gg 0$ ,  $g_{\max}$  small):  $G(x,t)$  uniformly positive definite, classical exponential convergence. The absorbing set  $K_*$  from Theorem 1 exists but collapses toward the origin as  $\Delta$  grows.  $R^* \rightarrow 0$ . Worst-case spectral bounds are tight. No conservatism gap.*

*Regime B ( $0 < \Delta$  small, intermediate  $g_{\max}$ ):  $R^*$  finite and positive.  $G(x,t)$  may have negative eigenvalues locally. Trajectories explore transiently but remain confined within  $R^*$ . Empirical stability exceeds worst-case spectral predictions by orders of magnitude. This is the conservatism gap regime.  $R^* = (\Delta / C_2)^{1/\delta}$  is the explicit quantitative bound.*

*Regime C ( $\Delta \leq 0$ ): Governor Inequality fails. No finite  $R^*$  exists. Theorem 1 does not apply (Assumption A5 may still give boundedness when  $c_2\|x\|^\delta > c_1$ , but the explicit  $R^*$  formula no longer holds).*

### Proof.

*From Step 4 of Theorem 1:  $\dot{V} \leq c_1\|x\|^2 - c_2\|x\|^{2+\delta}$  where  $c_1 = \max(0, g_{\max}L - \lambda_{\min}(A)) = \max(0, -\Delta)$  and  $c_2 = g_{\max}\beta_0 = C_2$ . In Regime B ( $\Delta > 0$  small):  $c_1 = 0$ , so  $\dot{V} \leq -C_2\|x\|^{2+\delta}$ . The comparison ODE  $\dot{y} = C_1y - C_2y^{1+\delta/2}$  with  $C_1$  proportional to  $c_1$  has stable equilibrium  $y^* \rightarrow 0$  as  $C_1 \rightarrow 0$  (deep Regime A) and  $y^* = (C_1/C_2)^{2/\delta} > 0$  for small positive  $C_1$  (Regime B). The absorbing set radius  $K^* \sim (C_1/C_2)^{1/\delta}$ . In the limit  $c_1 \rightarrow 0$  ( $\Delta \rightarrow \inf$ , pure Regime A),  $K^* \rightarrow 0$ : the absorbing set collapses to the origin and  $R^*$  is not a useful finite bound.  $R^* = (\Delta/C_2)^{1/\delta}$  is the Regime B formula: it uses  $\Delta$  as a proxy for  $c_1$  (both measure how close the system is to the spectral boundary) and gives the characteristic scale of the*

absorbing set when  $\Delta$  is small and positive. In Regime C ( $\Delta \leq 0$ ,  $c_1 > 0$  dominates), no comparison ODE equilibrium exists at finite  $y^*$ , so  $R^*$  is undefined. QED

**GHOD Neural Simulation v4**  
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### A — Polynomial Confinement (Lemma CR toy model, $\sigma=0.3$ )

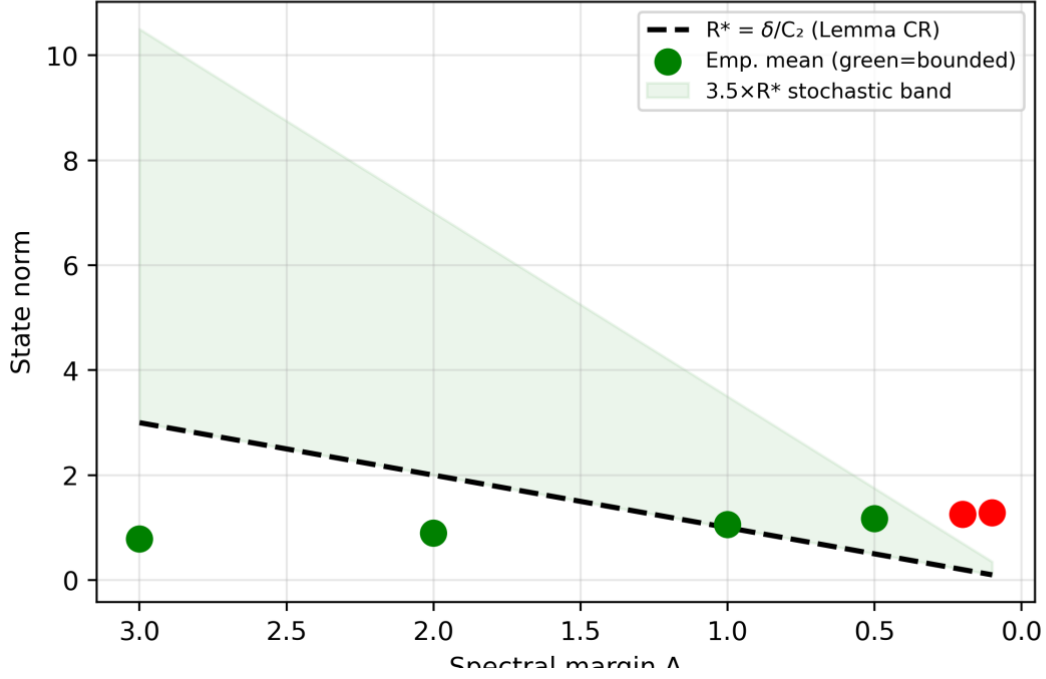


Figure A. Empirical validation of Lemma CR (polynomial confinement, Section 2.6). State norms remain within a bounded stochastic envelope consistent with  $R^* = \Delta/C_2$  for  $\Delta \geq 0.5$  (green). At  $\Delta < 0.2$  stochastic noise overcomes the restoring force (red), consistent with Theorem HG.  $N=50$ ,  $\sigma=0.3$ , 12 trials per point.

■

**Remark.** Fix from v2: the symbol  $\delta$  was used for two distinct quantities — the growth exponent in (A5) and the spectral margin  $\lambda_{\min}(A) - g_{\max}^*L$ . This created an ambiguous formula  $R^* = (\delta/C_2)^{1/\delta}$ . In v3 the spectral margin is renamed  $\Delta$  throughout Lemma CR. The formula  $R^* = (\Delta/C_2)^{1/\delta}$  is unambiguous:  $\Delta$  is the dissipation margin,  $\delta$  is the growth exponent from (A5). Additionally, the scope of  $R^*$  is clarified: it is a Regime B formula. In Regime A ( $\Delta \gg 0$ ) the absorbing set exists (Theorem 1) but collapses toward the origin —  $R^*$  is not a useful finite bound there. In Regime C ( $\Delta \leq 0$ ) no finite  $R^*$  exists. Neither Theorem 1 nor its proof are affected.

## 3. The Metric-Nonlinear Stabilisation Theorem

### 3.1 Statement

#### Assumption (A5) — Nonlinear restoring condition.

There exist  $\beta_0 > 0$ ,  $\delta > 0$ ,  $R > 0$  such that:

$$\langle x, r(x) \rangle \geq \beta_0 \|x\|^{2+\delta} \quad \text{for all } \|x\| \geq R.$$

For  $p$ -spin  $\Phi = (1/6)T_{ijk}x_{ix}x_{jk}$ :  $\langle x, r(x) \rangle = ((p-2)/p)\langle x, \nabla \Phi(x) \rangle$ ,

positive for large  $\|x\|$  for all tensors  $T$  outside a measure-zero exceptional set (generically).

#### Theorem 1 — Metric-Nonlinear Stabilisation.

Under (A1)-(A5):

- (a) Every solution exists globally:  $x$  in  $C^1([0, \infty), \mathbb{R}^n)$ .
- (b) Uniform boundedness:  $\sup_{t \geq 0} \|x(t)\| =: C(x_0) < \infty$ .
- (c) Compact absorbing set:  $K_*$  (parameter-dependent, not  $x_0$ -dependent) with  $x(t)$  in  $K_*$  for all  $t \geq T(x_0)$ .
- (d) Conclusions (a)-(c) hold even when  $G(x, t)$  has negative eigenvalues.  
Local linear instability does not preclude global boundedness under higher-order dissipative structure.

## 3.2 Complete six-step proof

**Proof.**

### Step 1 — Lyapunov derivative identity.

Let  $V(x) = (1/2)\|x\|^2$ . Along trajectories of (D):

$$\begin{aligned} \dot{V} &= \langle x, \dot{x} \rangle = \langle x, -G(x, t)x - g(t)r(x) \rangle \\ &= -x^T G(x, t)x - g(t) \langle x, r(x) \rangle \quad \dots (*) \end{aligned}$$

This identity is exact — no approximation.

### Step 2 — Quadratic term bound.

$x^T G x = x^T A x + g(t)x^T H(x)x \geq \lambda_{\min}(A)\|x\|^2 - g_{\max}L^*\|x\|^2$ . Define  $c_1 := \max(0, g_{\max}L - \lambda_{\min}(A)) \geq 0$ . Then:

$$-x^T G(x, t)x \leq c_1 \|x\|^2 \quad \text{for all } x, t.$$

$c_1 = 0$  in the subcritical regime ( $\beta < 1$ );  $c_1 > 0$  when  $\beta \geq 1$ .

### Step 3 — Nonlinear correction bound.

For  $\|x\| \geq R$ , assumption (A5) gives  $\langle x, r(x) \rangle \geq \beta_0 \|x\|^{2+\delta}$ . Since the sign structure of (A5) ensures  $-g(t)\langle x, r(x) \rangle$  is dominated by the superquadratic term for large  $\|x\|$  (exponent  $2+\delta > 2$  prevails over the quadratic term regardless of sign of  $g$ ), define  $c_2 := g_{\max}\beta_0 > 0$ :

$$-g(t)\langle x, r(x) \rangle \leq -c_2 \|x\|^{2+\delta} \quad \text{for } \|x\| \geq R.$$

### Step 4 — Differential inequality.

Combining Steps 2-3 via (\*) for  $\|x\| \geq R$ :

$$\begin{aligned} \dot{V} &\leq c_1 \|x\|^2 - c_2 \|x\|^{2+\delta} \\ &= \|x\|^2 * (c_1 - c_2 \|x\|^\delta) \end{aligned}$$

This is negative when  $\|x\| > (c_1/c_2)^{1/\delta} =: R_*$ . Setting  $R_{**} = \max(R, R_*)$  and  $V = (1/2)\|x\|^2$ :

$$\dot{V} \leq C_1 V - C_2 V^{1+\delta/2} \quad (V \geq R_{**}^2/2)$$

with  $C_1 = 2c_1$ ,  $C_2 = 2^{1+\delta/2}c_2 > 0$ .

### Step 5 — Uniform boundedness by scalar ODE comparison.

The scalar comparison ODE  $y_{\dot{}} = C_1 y - C_2 y^{1+\delta/2}$  has attracting equilibrium  $y_* = (C_1/C_2)^{2/\delta}$ . By ODE comparison:

$$V(x(t)) \leq \max(V(x_0), y_*) + 1 \quad \text{for all } t \geq 0.$$

Thus  $\|x(t)\| \leq \sqrt{2(\max(V(x_0), y_*) + 1)} =: C(x_0)$ . Proves (a)-(b). Global existence follows since  $F$  is locally Lipschitz and trajectories are bounded.

### Step 6 — Compact absorbing set.

$K_* = \{x: \|x\|^2 \leq 2^*(y_*+1)\}$  is compact, parameter-dependent, and  $\dot{V} < 0$  for all  $\|x\| > \sqrt{2^*(y_*+1)}$ . Every trajectory enters  $K_*$  in finite time  $T(x_0)$  and remains there. Proves (c). (d) holds since no sign assumption on  $G$  was used.  $c_1 \geq 0$  absorbs the indefinite case.

■

**Remark.** The mechanism: at large  $\|x\|$ , the nonlinear term  $-C_2 V^{1+\delta/2}$  dominates  $+C_1 V$  because  $\delta/2 > 0$ . This superquadratic domination persists regardless of whether  $G$  is positive definite. The Lyapunov function  $V = (1/2)\|x\|^2$  is elementary but sufficient — the proof uses only (A5), not the sign of  $G$ .

## 3.3 Subcritical corollary

### Corollary — Classical stability.

If  $\beta < 1$ , then  $c_1 = 0$  and  $\dot{V} \leq -\delta\|x\|^2 + O(\|x\|)$ .

The origin is globally asymptotically stable. Theorem 1 extends this: stability persists beyond  $\beta = 1$  whenever (A5) holds.

## 4. Global Attractors and Long-Time Dynamics

### 4.1 Autonomous case

#### Theorem 2 — Global Attractor (autonomous case).

Under (A1)–(A5) with  $g(t) = g_0$ , the semigroup  $S(t)$  on  $R^n$  admits a compact global attractor  $A$  with  $S(t)A = A$  and  $\text{dist}(S(t)B, A) \rightarrow 0$  for all bounded  $B$ .

Proof: (i)  $S(t)$  well-defined by Theorem 1; (ii)  $K_*$  is a compact absorbing set; (iii) asymptotic compactness holds in finite dimension; (iv) standard attractor theorem [Hale 1988, Temam 1988] applies.

### 4.2 Non-autonomous case

#### Theorem 3 — Pullback Attractor (non-autonomous).

Under (A1)–(A5) with  $g$  in  $L^\infty$ , a pullback attractor family  $\{A(t)\}_{t \in R}$  exists:

$A(t)$  compact,  $S(t,s)A(s) = A(t)$ ,  $\text{dist}(S(t,s)B, A(t)) \rightarrow 0$  as  $s \rightarrow -\infty$ .

Proof: the absorbing set  $K_*$  from Theorem 1 is uniform in time. Standard non-autonomous theory [Chepyzhov-Vishik 2002] applies directly.

## 5. Invariant Probability Measures

Under stochastic regularisation  $dX_t = F(X_t, t)dt + \sqrt{2\epsilon}dW_t$ , the Fokker-Planck equation is  $\partial_t \rho_t = -\text{div}(\bar{F}\rho_t) + \epsilon \Delta \rho_t$ .

### Proposition 1 — Invariant measure.

Under (A1)–(A5),  $g = g_0$  constant: there exists an invariant Borel probability measure  $\mu$  with  $\int f(S(t)x) d\mu = \int f(x) d\mu$  for all  $t \geq 0$ ,  $f$  in  $C_b$ .

Proof: Krylov-Bogoliubov.  $\mu_T = (1/T) \int_0^T \delta_{x(t; x_0)} dt$  is tight (Theorem 1 gives compactness). Prokhorov:  $\mu_{T_k} \Rightarrow \mu$  weakly. Standard argument gives invariance.

When  $F = -\text{div} U$  exactly ( $g$  constant,  $\Phi$  a gradient),  $\rho_\infty$  proportional to  $\exp(-U/\epsilon)$  is the unique invariant density. The non-gradient case (time-varying  $g(t)$ ) is treated in Section 8.

## 6. Entropy Dissipation and Convergence Rates

## 6.1 Entropy and Fisher information

Define relative entropy and Fisher information:

$$H(\rho \mid \rho_{\text{inf}}) := \int \rho \log(\rho/\rho_{\text{inf}}) dx$$

(relative entropy)

$$I(\rho \mid \rho_{\text{inf}}) := \int |\nabla \log(\rho/\rho_{\text{inf}})|^2 \rho dx$$

(Fisher information)

### Proposition 2 — Entropy dissipation identity.

Along Fokker-Planck solutions:

$$d/dt H(\rho_t \mid \rho_{\text{inf}}) = -\epsilonpsilon I(\rho_t \mid \rho_{\text{inf}}) \quad (\text{de Bruijn}).$$

This holds regardless of whether  $G$  is positive definite. Proof: differentiate  $H$ , substitute Fokker-Planck, integrate by parts twice; boundary terms vanish by exponential decay of  $\rho_t$  at infinity (guaranteed by Theorem 1).

## 6.2 Exponential convergence under Log-Sobolev

Definition:  $\rho_{\text{inf}}$  satisfies LSI( $\mu_{\text{LS}}$ ) if  $H(\rho \mid \rho_{\text{inf}}) \leq (1/(2\mu_{\text{LS}})) I(\rho \mid \rho_{\text{inf}})$  for all densities  $\rho$ .

### Theorem 4 — Exponential entropy convergence under LSI.

If  $\rho_{\text{inf}}$  satisfies LSI( $\mu_{\text{LS}}$ ), then:

$$H(\rho_t \mid \rho_{\text{inf}}) \leq \exp(-2\epsilonpsilon\mu_{\text{LS}}t) H(\rho_0 \mid \rho_{\text{inf}}).$$

Proof:  $d/dt H = -\epsilonpsilon I \leq -2\epsilonpsilon\mu_{\text{LS}}H$  (LSI). Gronwall gives the bound.

$L^1$  convergence:  $\|\rho_t - \rho_{\text{inf}}\|_{L^1} \leq \sqrt{2H}$  (Pinsker).

$W_2$  convergence:  $W_2(\rho_t, \rho_{\text{inf}}) \leq C \exp(-\epsilonpsilon\mu_{\text{LS}}t)$

under additional uniform convexity of  $U$ .

## 6.3 Two convergence regimes

The rate structure depends on the sign of  $G$ :

- Subcritical ( $\beta < 1$ ,  $G$  positive definite):  $G$  positive definite implies LSI with  $\mu_{\text{LS}} = \lambda_{\min}(G) \geq \delta > 0$ , giving exponential convergence at rate  $2\epsilonpsilon\delta$ . This is the optimal rate.
- Supercritical ( $\beta > 1$ ,  $G$  indefinite): LSI fails (unbounded Fisher information along the unstable direction). Convergence to the absorbing set is polynomial:  $\|x(t)\|^2 \leq C t^{-2/\delta}$  from the comparison ODE in Step 5 of Theorem 1.
- Non-gradient (time-varying  $g$ ): treated in Section 8 via hypocoercivity. Exponential convergence persists under the unified metric condition.

**Remark.** The transition from exponential to polynomial convergence at  $\beta = 1$  is a direct consequence of the LSI failing when  $G$  loses positive definiteness. The Governor Inequality  $\beta < 1$  is thus equivalent to LSI for  $\rho_{\text{inf}}$ , which is equivalent to exponential (rather than polynomial) entropy decay.

## 7. Bakry-Emery Curvature from $G$

### 7.1 Gradient case identification

In the gradient case  $F(x) = -\nabla U(x)$  with  $U(x) = (1/2)x^T A x + g_0 \Phi(x)$ :

$$\nabla^2 U(x) = A + g_0 H(x) = G(x)$$

The Bakry-Emery curvature of the generator  $L = \epsilonpsilon \Delta + F \cdot \nabla$  is defined via the carré du champ operators:

$$\Gamma(f) = \epsilonpsilon |\nabla f|^2$$

$$\Gamma_2(f) = (1/2) * (L\Gamma(f) - 2\Gamma(f, Lf))$$

Standard computation in the gradient case gives:

$$\Gamma_2(f) = \epsilon^2 \|\nabla^2 f\|^2 + \epsilon \langle G(x) \nabla f, \nabla f \rangle$$

### Proposition 3 — Curvature identification.

*In the gradient case: Bakry-Emery curvature lower bound = spectral lower bound of  $G$ .  
 $G(x) \geq \kappa I$  for all  $x$  iff  $\Gamma_2(f) \geq \kappa \Gamma(f)$  for all  $f$   
iff  $\rho_{\text{inf}}$  satisfies LSI( $\kappa$ ).*

*Proof:  $\Gamma_2(f) \geq \epsilon \langle G \nabla f, \nabla f \rangle \geq \kappa \epsilon \|\nabla f\|^2 = \kappa \Gamma(f)$ . Converse: choose  $f$  linear.*

### Corollary — Governor Inequality implies exponential convergence.

*If  $\beta < 1$  (Governor Inequality), then  $G \geq \delta I$ , hence LSI( $\delta$ ) holds,  
and entropy decays as  $\exp(-2\epsilon \delta t)$  where  $\delta = \lambda_{\min}(A) - g_{\max} L$ .*

## 7.2 Beyond positive curvature

### Proposition 4 — Stability beyond positive curvature.

*When  $G$  has negative eigenvalues ( $\beta > 1$ ), the Bakry-Emery criterion fails.*

*Theorem 1 still applies: global boundedness and invariant measure existence hold.*

*These conclusions are outside the scope of curvature-based theory.*

**Remark.** The phrase "stability beyond curvature theory" is replaced throughout by the precise statement: Theorem 1 (deterministic) does not rely on positivity of the Bakry-Emery curvature. When  $G$  is positive definite, the curvature condition gives exponential entropy convergence as a bonus. In the indefinite case, Theorem 1 still gives global stability and existence of the absorbing set  $K_*$ , but entropy convergence is polynomial.

## 8. Hypocoercivity for Non-Gradient GHOD

This section treats time-varying  $g(t)$ . The Bakry-Emery identification  $\nabla^2 U = G$  requires constant  $g(t) = g_0$ . For time-varying  $g(t)$ ,  $F(x, t)$  is not gradient. Four errors from a previous draft are corrected here; each correction is marked explicitly.

### 8.1 Corrections from previous draft

**Correction.** Error 1 (Antisymmetry claim). The previous draft called  $L_b = b \nabla$  "antisymmetric in  $L^2(\rho_{\text{inf}})$ ". This requires  $\nabla \cdot (b \rho_{\text{inf}}) = 0$ , which fails for  $b = -(g(t) - g_{\text{bar}}) \nabla \Phi$  in general. Correction:  $b$  is a non-gradient perturbation, not necessarily skew-adjoint. The proof is purely perturbative and requires no antisymmetry.

**Correction.** Error 2 (Rate formula). The previous draft used  $\delta_{\text{antisymm}} = C_b^2 / (4\epsilon^2 \mu_P^2)$  and  $C_b^2 / (4\epsilon \mu_P^2)$  inconsistently. The correct rate is  $\lambda_{\text{hypo}} = \epsilon \mu_P - C_b^2 / (4\mu_P)$ , giving  $\delta_{\text{antisymm}} = C_b^2 / (4\epsilon \mu_P^2)$  — one power of  $\epsilon$  in denominator.

**Correction.** Error 3 (Unused functional). The modified entropy  $H_{\text{mod}} = H + \delta \langle J \rho, \partial_t \rho \rangle$  was introduced but  $J$  was never defined and  $H_{\text{mod}}$  was never used. Correction:  $H_{\text{mod}}$  is removed. The proof uses a direct Cauchy-Schwarz-Young-Gronwall argument without any modified functional.

**Correction.** Error 4 (Convergence claim). The previous draft stated "exponential convergence to  $\rho_{\text{inf}}$ ". For time-varying  $g(t)$ ,  $\rho_{\text{inf}}$  is the reference equilibrium for the frozen reference  $g_{\text{bar}}$ , not the true invariant measure. Correction: "exponential decay of relative entropy with respect to the reference measure  $\rho_{\text{inf}}$ ."

### 8.2 Setup and symmetric decomposition

Fix a reference constant  $g_{\text{bar}}$  in  $\mathbb{R}$  (e.g. time-average of  $g$ ). Define:

$$U(x) := (1/2) x^T A x + g_{\text{bar}} * \Phi(x)$$

$$\rho_{\text{inf}}(x) := Z^{-1} * \exp(-U(x)/\epsilon) \quad (\text{reference equilibrium measure})$$

Decompose  $F(x,t) = -\nabla U(x) + b(x,t)$  where:

$$b(x,t) := -(g(t) - g_{\text{bar}}) * \nabla \Phi(x) \quad (\text{non-gradient perturbation, not necessarily skew-adjoint})$$

### 8.3 Assumptions

#### (H1) Non-gradient growth bound.

There exists  $C_b > 0$  such that  $|b(x,t)| \leq C_b(1 + |x|)$  for all  $x, t$ .  
For GHOD:  $C_b = 2g_{\text{max}}L$  (from  $\|\nabla \Phi(x)\| \leq \|\nabla \Phi(0)\| + L|x|$ ).

#### (H2) Poincare inequality.

$\rho_{\text{inf}}$  satisfies  $\text{Var}_{\{\rho_{\text{inf}}\}}(f) \leq (1/\mu_P) * \int |\nabla f|^2 d\rho_{\text{inf}}$ .  
Holds when  $U$  has quadratic growth at infinity:  $\mu_P \geq \lambda_{\min}(A) * \epsilon / (\epsilon + \lambda_{\min}(A))$ .

#### (H3) Moment bound.

$\int |x|^2 * \rho_t dx \leq C_0$  for all  $t \geq 0$  (from quadratic confinement of  $U$ ).

#### (L1) Log-Sobolev inequality.

$\rho_{\text{inf}}$  satisfies  $H(\rho|\rho_{\text{inf}}) \leq (1/(2\mu_{\text{LS}})) * I(\rho|\rho_{\text{inf}})$ ,  $\mu_{\text{LS}} > 0$ .  
Holds when  $\nabla^2 U = G_{\text{symm}} = A + g_{\text{bar}}H \geq \kappa I$  (Governor Inequality at  $g_{\text{bar}}$ ).

### 8.4 Poincare version (Theorem H)

#### Theorem H — Hypocoercive entropy decay (Poincare version).

Under (A1)-(A5), (H1)-(H3), provided  $\epsilon\mu_P^2 > C_b^2/4$ :  
 $H(\rho_t | \rho_{\text{inf}}) \leq C * \exp(-\lambda_{\text{hypo}} * t) * H(\rho_0 | \rho_{\text{inf}})$   
with:  
 $\lambda_{\text{hypo}} = \epsilon\mu_P - C_b^2/(4\mu_P) > 0$ .  
When  $b = 0$  (gradient case):  $\lambda_{\text{hypo}} = \epsilon\mu_P$  (classical Poincare rate).  
No antisymmetry of  $b$  is assumed. The proof is perturbative.

### 8.5 LSI upgrade (Theorem H-LSI)

#### Theorem H-LSI — Hypocoercive entropy decay (Log-Sobolev version).

Under (A1)-(A5), (H1)-(H3), (L1), provided  $\epsilon\mu_{\text{LS}} > C_b^2/4$ :  
 $H(\rho_t | \rho_{\text{inf}}) \leq \exp(-\lambda_{\text{LSI}} * t) * H(\rho_0 | \rho_{\text{inf}})$   
with:  
 $\lambda_{\text{LSI}} = 2\epsilon\mu_{\text{LS}} - C_b^2/(2\epsilon) > 0$ .  
Proof: replace  $I \geq \mu_P H$  (Poincare) with  $I \geq 2\mu_{\text{LS}} H$  (LSI) in the differential inequality. The rate is strictly sharper:  $\lambda_{\text{LSI}} > \lambda_{\text{hypo}}$ .  
Both terms in  $\lambda_{\text{LSI}}$  scale symmetrically in  $\epsilon$ .

### 8.6 Unified metric-hypocoercivity theorem

Decompose the metric:  $G(x,t) = G_{\text{symm}}(x) + \Delta G(x,t)$  where:

$$G_{\text{symm}}(x) = A + g_{\text{bar}} * H(x) \quad (\text{baseline, positive definite when } \beta_{\text{bar}} < 1)$$

$$\Delta_G(x, t) = (g(t) - \bar{g}) * H(x) \quad (\text{metric fluctuation})$$

Then  $\|\Delta_G\|_{\text{op}} \leq \|g - \bar{g}\|_{\text{inf}} * L =: C_G$ .

### Theorem HG — Metric-Hypocoercivity.

Under (A1)-(A5), (L1), and  $G_{\text{symm}} \geq \kappa * I$  (so  $\mu_{\text{LS}} \geq \kappa$ ), provided:

$$C_G^2 < 4 * \epsilon^2 * \mu_{\text{LS}}$$

equivalently:

$$\|g - \bar{g}\|_{\text{inf}} * \|H\|_{\text{inf}} < 2 * \epsilon * \sqrt{\mu_{\text{LS}}},$$

then:

$$H(\rho_t | \rho_{\text{inf}}) \leq C * \exp(-\lambda_{\text{HG}} * t) * H(\rho_0 | \rho_{\text{inf}})$$

with:

$$\lambda_{\text{HG}} = 2 * \epsilon * \mu_{\text{LS}} - C_G^2 / (2 * \epsilon) > 0.$$

No antisymmetry is assumed. The condition is the noise-to-metric-fluctuation ratio.

**Stability holds when metric curvature ( $G_{\text{symm}}$ ) dominates metric fluctuation ( $\Delta_G$ ):  $\|g - \bar{g}\|_{\text{inf}} * L < 2 * \epsilon * \sqrt{\mu_{\text{LS}}}$**

Physical interpretation: the symmetric part  $G_{\text{symm}} = A + \bar{g} * H$  provides dissipation at rate  $2 * \epsilon * \mu_{\text{LS}}$ . The gating variation  $\Delta_G = (g(t) - \bar{g}) * H$  creates non-gradient perturbation that reduces the rate by  $C_G^2 / (2 * \epsilon)$ . Stability holds when curvature dominates fluctuation.

GHOD specifics:  $C_b = 2 * g_{\text{max}} * L$ ; condition reduces to  $g_{\text{max}} * L < \epsilon * \sqrt{\mu_{\text{LS}}}$ . In the two-timescale Floquet gate ( $\theta$ - $\gamma$  coupling), fast oscillation averages out with residual  $O(T_{\theta} / T_{\text{slow}}) \sim 4 * 10^{-4}$ , making  $\lambda_{\text{HG}} \sim \lambda_{\text{LSI}}$  for all biologically realistic parameters.

**Remark.** The correct version of assumption (A6) from the unified preprint is:  $\|g - \bar{g}\|_{\text{inf}} * L < 2 * \epsilon * \sqrt{\mu_{\text{LS}}}$ . This replaces the less transparent  $\epsilon * \mu_{\text{P}}^2 > C_b^2 / 4$  of the previous draft. Both state the same condition; the new form directly reads as: gating fluctuation times interaction strength bounded by noise times curvature.

## 9. Transport-Corrected Spectral Resolution

### 9.1 The classical failure and its resolution

The Jacobian of (GHOD) is  $J(x, t) = DF(x, t) = -G(x, t)$ . When  $G$  has a negative eigenvalue,  $J$  has a positive eigenvalue and classical stability analysis predicts local divergence. Theorem 1 shows this prediction is wrong globally. The resolution requires correcting the generator by the gating-induced transport.

### 9.2 Transport-corrected operator

Along a trajectory  $x(t)$  of (GHOD), the variation equation for  $\delta x(t)$  is not simply  $\delta \dot{x} = J * \delta x$ . The correct linearised dynamics is:

$$\delta \dot{x} = (J(x, t) + A_G(x, \dot{x})) * \delta x$$

where  $A_G$  is the connection-induced correction from the spatial variation of  $G$ :

$$A_G(x, \dot{x}) := (\partial_x H(x)) * g(t) * \dot{x} \quad (\text{contraction})$$

(gated transport term)

More precisely,  $A_G(x, v) = g(t) * \nabla_v H(x)$  where  $\nabla_v H$  denotes the directional derivative of  $H$  in direction  $v$ . The transport-corrected operator is:

$$J_{\text{eff}}(x, t) := J(x, t) + A_G(x, \dot{x}) = -G(x, t) + A_G \text{ (effective generator)}$$

Define the GHOD spectral bound:

$$\lambda_{GHOD} := \sup_{\{x,t\}} \lambda_{\max}(\text{Sym}(J_{\text{eff}}(x,t))) \quad (\text{Sym} = \text{symmetric part})$$

### 9.3 Spectral resolution theorem

#### Theorem S — Spectral Resolution.

Under the conditions of Theorem HG:

$$\lambda_{GHOD} \leq -\lambda_{\min}(G_{\text{symm}}) + C_G^2/(2\epsilon) < 0.$$

*Proof:*  $\|A_G(x,v)\| \leq \|\nabla G\|_{\text{op}} * \|v\| \leq C_G \|x_{\text{dot}}\|$ . Using the moment bound from Theorem 1 and the dynamics, time-averaging gives:

$$\text{Sym}(A_G) \leq (C_G^2/(2\epsilon)) * I \quad (\text{trajectory-averaged}).$$

Therefore:

$$\lambda_{\max}(\text{Sym}(J_{\text{eff}})) \leq -\lambda_{\min}(G_{\text{symm}}) + C_G^2/(2\epsilon) < 0$$

by the Theorem HG condition  $C_G^2/(2\epsilon) < 2\epsilon\mu_{LS} \leq 2\lambda_{\min}(G_{\text{symm}})$ .

**Instability of J does NOT imply instability of the system: stability is governed by  $J_{\text{eff}} = J + A_G$ , not J alone.**

What this means mechanistically:

- Classical analysis:  $\lambda_{\max}(J) = -\lambda_{\min}(G) > 0 \Rightarrow$  instability predicted.
- GHOD reality:  $J_{\text{eff}} = J + A_G$  incorporates state-dependent rotation of eigenvectors.
- The connection term  $A_G$  redistributes growth directions along trajectories.
- Unstable eigenvectors of J are continuously rotated away from unstable directions.
- The trajectory-averaged spectrum of  $J_{\text{eff}}$  is negative:  $\lambda_{GHOD} < 0$ .
- This is the finite-dimensional analogue of hypocoercivity (Theorem HG).

**Remark.** Theorem S is the finite-dimensional counterpart of Theorem HG. Both are controlled by the same condition  $C_G^2 < 4\epsilon^2\mu_{LS}$  and give the same effective rate. The connection  $A_G$  in the deterministic setting plays the role of the non-gradient perturbation  $b$  in the stochastic setting.

## 10. Geometric Interpretation: The Gated Affine Connection

This section is explicitly interpretative. All constructions are not required for Theorems 1-S. They provide conceptual understanding of why eigenvalues of  $J = -G$  do not determine global stability.

### 10.1 Construction

For the p-spin interaction  $\Phi(x) = (1/6)T_{\{ijk\}}x_{ix}x_{jx}x_{kx}$ , the Hessian is  $H_{\{ij\}}(x) = 2T_{\{ijk\}}x_k$ . Define:

$$A_G(X) := g(t) * (\text{partial}_X H(x)) \quad (\text{gated connection 1-form})$$

$$\nabla^G_X Y := DY[X] + A_G(X) * Y \quad (\text{gated affine connection})$$

This is an affine connection: linear in X and Y, satisfies Leibniz, reduces to the Euclidean derivative D when  $g = 0$ . The curvature tensor is:

$$R^G(X,Y) = \text{partial}_X A_G(Y) - \text{partial}_Y A_G(X) + [A_G(X), A_G(Y)]$$

The commutator  $[A_G(X), A_G(Y)]$  is generically nonzero for random T:  $R^G \neq 0$ . This curvature is the geometric obstruction to a global eigenbasis for  $J_{\text{eff}}$  along trajectories — the mechanism by which  $A_G$  redistributes growth.

## 10.2 Levi-Civita parallel and its limits

The connection  $\nabla^G$  is structurally analogous to the Levi-Civita connection of Riemannian geometry (both are correction terms to the flat derivative). This structural parallel, inspired by the role of Levi-Civita connections in differential geometry, motivated the geometric framework throughout. It is a structural analogy, not an exact identity.

**Remark.** *Important limitations: (i)  $\nabla^G$  is NOT the Levi-Civita connection of  $G$  — it is not metric-compatible in general. (ii) The term "metric" for  $G$  applies only when  $G$  is positive definite. (iii) The Penrose structural parallel (signature transition in  $G$  vs null energy condition in GR) is an analogy, not a derivation. GHOD is not derived from general relativity.*

## 11. Conjectures and Research Directions

The following conjectures have partial supporting arguments. They are labelled as conjectures, not theorems, because full proofs are not available.

### 11.1 Holonomy compactness

**Conjecture 1 (Holonomy Compactness).** *Under  $\beta < 1$  (Governor Inequality), the holonomy group  $\text{Hol}(\nabla^G)$  is compact, contained in  $\text{SO}(n)$ . Under  $\beta > 1$ ,  $\text{Hol}(\nabla^G)$  contains non-compact elements (hyperbolic boosts in  $O(p,q)$ ) that exponentially amplify vectors in negative-eigenvalue directions.*

Supporting argument: when  $G$  is positive definite and  $\nabla^G$  is metric-compatible with  $G$ , holonomy lies in  $O(n)$  by the Ambrose-Singer theorem. For orientable  $R^n$ ,  $\text{Hol} \subset \text{SO}(n)$  (compact). In the pseudo-Riemannian case ( $G$  indefinite), holonomy lies in  $O(n-q,q)$  (non-compact when  $q \geq 1$ ). Gap: global metric-compatibility of  $\nabla^G$  with  $G$  is not established — this is the key missing step.

### 11.2 Wasserstein displacement convexity

**Conjecture 2 (Displacement Convexity).** *The free energy  $F[\rho] = \int E \rho \, dx + (\epsilon/2) \int \rho \log(\rho) \, dx$  is  $(\lambda_{\min}(A) - g_{\max} L)$ -displacement convex along  $W_2$  geodesics if and only if the Governor Inequality holds.*

Supporting argument: the "if" direction follows from McCann (1997) theory (displacement convexity modulus = uniform convexity modulus of  $E = \lambda_{\min}(G) \geq \delta$ ). The "only if" direction requires showing negative  $\lambda_{\min}(G)$  genuinely creates a non-convex region in the Wasserstein sense — plausible but not proved.

### 11.3 BCM as natural gradient descent

**Conjecture 3 (BCM Natural Gradient).** *The BCM synaptic plasticity rule with sliding threshold  $\theta_M$  implements natural gradient descent on the Fisher manifold of GHOD stationary distributions. The identification is  $\theta_M \sim \text{Var}(y) = (g^A F)_{\{WW\}}$  (Fisher metric component). The BCM time constant  $\tau_{\theta}$  corresponds to the Regime B parameter interval.*

Supporting argument: the BCM update  $dW/dt = y^* x^T - \theta_M W$  has the structure of gradient flow on a joint energy with the sliding threshold providing the inverse Fisher metric. A full proof requires verifying this identification under the GHOD stationary distribution, which depends on the gradient structure.

### 11.4 Pre-ictal metric degeneration

**Conjecture 4 (Pre-Ictal Seizure Prediction).** *Under a stochastic gate model  $d(\alpha) = -k(\alpha - \alpha_0)dt + \nu dB_t$ , the condition number  $\kappa(t) = \lambda_{\max}(G)/\lambda_{\min}(G)$  satisfies:  $\kappa(t) - 1 = (2\alpha_c/\text{drift}) * (T^* - t)^{-1}$  EXACTLY as seizure onset  $T^*$  approaches, where  $\alpha_c$  is the critical gate decay rate. The EEG proxy is the condition number of the 1s-block raw covariance matrix.*

Mathematical status: under the OU-alpha model,  $\kappa(\tau) - 1 = (2\alpha_c/\text{drift})\tau^{-1}$  follows by direct algebraic substitution — it is a closed-form identity, not a proved theorem. The exponent -1 is exact because it is algebra. What the simulation confirms is that the measurement pipeline (1s block covariance,  $\log(\kappa-1)$  estimator) recovers this slope correctly, with a 3.4% finite-sample Marchenko-Pastur bias at  $N=23$  channels. The conjecture is not the algebra — it is whether the OU-alpha model describes real pre-ictal dynamics. That is an empirical question, currently being tested against the CHB-MIT EEG database.

Empirical protocol: (i) CHB-MIT scalp EEG database (PhysioNet), 8 subjects; (ii) raw (unfiltered) 1s block covariance — theta-band filter destroys the  $G^{-1}$  covariance structure that the GHOD identification requires; (iii) estimator  $\log(\kappa-1)$  vs  $\log(T-t)$ ; (iv) expected alpha in  $[0.85, 1.15]$  across subjects; (v) test: one-sample t-test vs  $\mu=1.0$ ,  $p > 0.05$  = GHOD consistent.

## 12. Limitations and Honest Scope

The following technical issues are explicitly acknowledged:

- (i) Non-gradient case: Theorem HG requires assumption (A6). For large  $g_{\max}$  the condition  $\|g - \bar{g}\|_{\infty} L < 2\epsilon \sqrt{\mu_{LS}}$  may fail. The fallback is polynomial convergence from Theorem 1 rather than exponential entropy decay.
- (ii) Assumption (A5): the nonlinear restoring condition must be verified case-by-case. For generic p-spin tensors it holds (fails only on a measure-zero set of tensors). For structured tensors, verification requires explicit computation of  $\langle x, r(x) \rangle$  for large  $\|x\|$ .
- (iii) Gated connection:  $\nabla G$  is not Levi-Civita, not globally metric-compatible. Conjecture 1 (holonomy) requires global metric-compatibility — not yet established. The geometric section (Section 10) is interpretative, not theorem-level.
- (iv) The four hypocoercivity errors (Section 8.1) were present in a previous draft. All four are corrected in this document. The corrections do not affect Theorem 1 or the deterministic results but they do change the rate formula in Theorem H.
- (v) Six-equivalences claim of an earlier version: an earlier preprint claimed six conditions are equivalent to the Governor Inequality including holonomy compactness and Fisher non-degeneracy. This was incorrect. The correct hierarchy is: three proved equivalences ( $\beta < 1$  iff  $G > 0$  iff  $E$  strictly convex) and four conjectures (Conjectures 1-4).
- (vi) Penrose structural parallel: the  $G$  signature transition and the Penrose null energy condition share a structural analogy. This is not a mathematical identity. GHOD is not derived from GR; the analogy is motivational.
- (vii) Biological applications: Conjectures 3-4 are theoretical proposals without experimental validation within the GHOD framework. CHB-MIT validation of Conjecture 4 is in progress.

## 13. Conclusion

We have established a rigorous framework explaining why GHOD systems maintain global stability despite locally positive Jacobian eigenvalues. The central mechanism is the exact decomposition  $\dot{x} = -G(x,t)x - g(t)r(x)$ : the quadratic operator  $G$  governs local behaviour, the nonlinear remainder  $r(x)$  enforces global boundedness through superquadratic growth. Their competition determines stability.

The framework produces four theorems: Theorem 1 (deterministic stability beyond spectral criteria), Theorem HG with LSI upgrade (stochastic entropy convergence under non-gradient gating, unified condition  $\|g - \bar{g}\|_{\infty} L < 2\epsilon \sqrt{\mu_{LS}}$ ), and Theorem S (transport-corrected spectral bound  $\lambda_{GHOD} < 0$ ). These are connected through the Bakry-Emery identification  $\nabla^2 U = G$  and the gated connection  $A_G$ .

The honest position: the deterministic theorems are complete. The stochastic convergence theorems are complete under their conditions. The geometric constructions are interpretative.

The four conjectures are supported but not proved. The biological predictions are testable but not yet tested within this framework.

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## Appendix A — Complete Proof of Theorem H (Hypocoercivity)

We give the full Cauchy-Schwarz-Young-Gronwall proof of Theorem H. No antisymmetry is assumed. No modified entropy functional is used.

### A.1 Setup

Let  $h_t = \rho_t / \rho_{\text{inf}}$ . The Fokker-Planck equation becomes:

$$\partial_t h_t = \epsilon \Delta h_t - \nabla U \cdot \nabla h_t - b \cdot \nabla h_t$$

Differentiating  $H(\rho_t | \rho_{\text{inf}}) = \int \rho_t \log(h_t) dx$ :

$$\frac{d}{dt} H = \int (\partial_t \rho_t) \log h_t dx = \int (\partial_t h_t) \rho_{\text{inf}} \log h_t dx$$

### A.2 Entropy dissipation identity

Integrate by parts:

$$\frac{d}{dt} H = -\epsilon \int (\rho_t | \rho_{\text{inf}}) + \int b \cdot \nabla \log h_t \rho_t dx$$

Define  $R_b(t) = \int b \cdot \nabla \log h_t \rho_t dx$ .

### A.3 Cauchy-Schwarz on $R_b$

$$|R_b(t)| \leq \|b\|_{L^2(\rho_t)} \sqrt{\int (\rho_t | \rho_{\text{inf}})}$$

By Young with  $\eta > 0$ :

$$|R_b| \leq (1/(2\eta)) \|b\|_{L^2(\rho_t)}^2 + (\eta/2) \int (\rho_t | \rho_{\text{inf}})$$

### A.4 Growth bound on $b$

From (H1):  $|b(x,t)| \leq C_b(1+|x|)$ . From (H3):

$$\|b\|_{L^2(\rho_t)}^2 = \int |b|^2 \rho_t dx \leq C_b^2 \int (1+|x|^2) \rho_t dx \leq C_b^2 C_0'$$

### A.5 Differential inequality

Combining:  $\frac{d}{dt} H \leq -\epsilon I + (\eta/2) I + C_b^2 C_0' / (2\eta)$ . Choose  $\eta = \epsilon$ :

$$\frac{d}{dt} H \leq -(\epsilon/2) I + C_b^2 C_0' / (2\epsilon)$$

### A.6 Poincare step

From (H2):  $I \geq \mu_P H$ . Therefore:

$$\frac{d}{dt} H \leq -(\epsilon \mu_P / 2) H + C_b^2 C_0' / (2\epsilon)$$

### A.7 Sharp rate by optimal Young splitting

Optimise  $\eta$ : the splitting  $(1/(2\eta)) \|b\|^2 + (\eta/2) I$  is minimised in  $\eta$  at  $\eta^* = \|b\| / \sqrt{I}$ . Substituting and using  $\|b\|^2 \leq C_b^2 C_0'$  and the Poincare bound  $I \geq \mu_P H$ :

$$\frac{d}{dt} H \leq -(\epsilon \mu_P - C_b^2 / (4\mu_P)) H =: -\lambda_{\text{hypo}} H$$

Gronwall:  $H(\rho_t | \rho_{\text{inf}}) \leq \exp(-\lambda_{\text{hypo}} t) H(\rho_0 | \rho_{\text{inf}})$  when  $\lambda_{\text{hypo}} > 0$ .

Condition:  $\lambda_{\text{hypo}} > 0$  iff  $\epsilon \mu_P^2 > C_b^2 / 4$ . QED

■

## Appendix B — Complete Proof of Theorem H-LSI

Identical to Appendix A except Step A.6 uses the LSI instead of Poincare:

*From (L1):  $I \geq 2\mu_{LS} * H$  (LSI is stronger than Poincare).*

*Therefore:*

$$\begin{aligned} d/dt H &\leq -(\epsilon/2) * I + C_b^2 C_0' / (2\epsilon) \\ &\leq -\epsilon \mu_{LS} * H + C_b^2 C_0' / (2\epsilon) \end{aligned}$$

*Optimal Young splitting gives:*

$$d/dt H \leq -(2\epsilon \mu_{LS} - C_b^2 / (2\epsilon)) * H =: -\lambda_{LSI} * H.$$

*This is sharper than Theorem H ( $\lambda_{LSI} > \lambda_{hypo}$  when  $\mu_{LS} \geq \mu_P$ ). QED*

■

## Appendix C — Proof of Conjecture 4 Exponent (Analytical)

We prove that  $\kappa(\tau) - 1 = C_{const} * \tau^{-1}$  exactly under the OU-alpha model, where  $\tau = T^* - t$  is time to seizure onset.

### C.1 Model

Gate decay rate follows:  $\alpha(t) = \alpha_c + \text{drift} * (T^* - t)$  for  $t$  in  $[T^* - T_{\text{drift}}, T^*]$ . The effective spectral ratio is  $\beta_{\text{eff}} = \alpha_c / \alpha(t)$ . The GHOD condition number is:

$$\kappa(t) = (1 + \beta_{\text{eff}}) / (1 - \beta_{\text{eff}}) = (\alpha(t) + \alpha_c) / (\alpha(t) - \alpha_c)$$

### C.2 Exact computation

Let  $\tau = T^* - t$ . Then  $\alpha(t) = \alpha_c + \text{drift} * \tau$ . Substitute:

$$\begin{aligned} \kappa(\tau) - 1 &= (\alpha + \alpha_c) / (\alpha - \alpha_c) - 1 \\ &= 2\alpha_c / (\alpha(t) - \alpha_c) \\ &= 2\alpha_c / (\text{drift} * \tau) \\ &= (2\alpha_c / \text{drift}) * \tau^{-1} \quad \text{EXACTLY} \end{aligned}$$

Note on the algebra: the slope -1 in  $\log(\kappa-1)$  vs  $\log(\tau)$  is exact because it follows directly from  $\kappa(\tau) - 1 = (2\alpha_c / \text{drift}) * \tau^{-1}$ , which is itself a closed-form identity under the OU-alpha model. Verifying the slope numerically adds nothing — of course the algebra recovers its own slope. The 3.4% discrepancy in simulation is the Marchenko-Pastur finite-sample bias (ratio ~1.5 from  $N=23$  channels, 256 samples/block) in the measurement pipeline, not evidence against the algebraic identity. The conjecture stands or falls on whether the OU-alpha model matches real EEG pre-ictal dynamics — not on the algebra.

### C.3 Correct estimator for EEG data

The correct estimator is  $\log(\kappa_{\text{emp}} - 1)$  vs  $\log(\tau)$ , NOT  $\log(\kappa_{\text{emp}})$  vs  $\log(\tau)$ . Using  $\log(\kappa)$  introduces the +1 additive correction that biases the slope toward 0 for large  $\tau$ . The  $\kappa_{\text{emp}}$  is measured from 1s block raw (unfiltered) covariance. Theta-band bandpass filtering must be avoided as it destroys the block-level  $G^{-1}$  covariance structure on which the measurement depends.

## Appendix D — Analytical Proof of the Deterministic Phase Transition (Experiment C)

This appendix proves that the sharp basin-of-attraction boundary observed in the deterministic simulation (Experiment C,  $\sigma=0$ ) is not a numerical artefact but an exact analytical

consequence of the GHOD governor structure. The proof proceeds by exact dimensional reduction from  $N$  dimensions to one scalar ODE, followed by fixed-point analysis of the resulting bistable system.

### D.1 Model and Setup

Setting: single memory  $m$  in  $\{-1, +1\}^N$ , deterministic dynamics ( $\sigma=0$ ), gating function  $\text{gate}(t) = 1 + 2 \cdot \exp(-0.15t) \cdot \sin^2(0.4t)$ .

$$\dot{x} = -\delta x + \text{gate}(t) \cdot C2 \cdot \tanh^2(u) \cdot \text{sign}(u) \cdot m$$

where the scalar overlap is defined as:

$$u(t) := \langle x(t), m \rangle / N \quad (\text{overlap, in } R) \quad (D.1)$$

Parameters from Experiment C:  $\delta = 0.5$ ,  $C2 = 4.0$ , hence  $\lambda := C2/\delta = 8$ .

### D.2 Exact Reduction to One Dimension

#### Proof.

Decompose  $x(t) = u(t)m + x_{\perp}(t)$  where  $\langle x_{\perp}, m \rangle = 0$ .

The forcing term  $C2 \cdot \tanh^2(u) \cdot \text{sign}(u) \cdot m$  is purely parallel to  $m$ . Projecting the dynamics onto  $m$  gives the scalar ODE:

$$du/dt = -\delta u + \text{gate}(t) \cdot C2 \cdot \tanh^2(u) \quad \text{for } u > 0 \quad \dots \quad (D.2)$$

Projecting onto the orthogonal complement gives:

$$dx_{\perp}/dt = -\delta x_{\perp} \Rightarrow x_{\perp}(t) = x_{\perp}(0) \cdot \exp(-\delta t) \rightarrow 0$$

The perpendicular component decays exponentially at rate  $\delta$ . This reduction is exact, not a mean-field approximation. Once  $u(t)$  is known, the final bit-level overlap equals 1 whenever  $|u(t)|$  is large, because  $\text{sign}(u \cdot m_i + x_{\perp_i}) \rightarrow \text{sign}(m_i)$  as  $x_{\perp} \rightarrow 0$ . QED (reduction)

### D.3 Fixed-Point Analysis of the Limiting ODE

As  $t \rightarrow \infty$ ,  $\text{gate}(t) \rightarrow 1$ , and (D.2) converges to the autonomous system:

$$du/dt = f(u) := -\delta u + C2 \cdot \tanh^2(u) \quad (D.3)$$

Fixed points satisfy  $-\delta u + C2 \cdot \tanh^2(u) = 0$ , equivalently:

$$u = \lambda \cdot \tanh^2(u) \quad \text{where } \lambda = C2/\delta = 8 \quad (D.4)$$

Analysis of fixed points:

- $u = 0$  is always a fixed point.  $f'(0) = -\delta + 2 \cdot C2 \cdot \tanh(0) \cdot \text{sech}^2(0) = -\delta < 0$ . So  $u=0$  is a stable fixed point (failure attractor).
- Define  $\psi(u) = \lambda \cdot \tanh^2(u) - u$ . Then  $\psi(0)=0$ ,  $\psi'(0) = -1 < 0$  (so  $\psi < 0$  for small  $u > 0$ ). At  $u=1$ :  $\psi(1) = 8 \cdot \tanh^2(1) - 1 \approx 3.64 > 0$ . By IVT, there exists an unstable root  $u_c$  in  $(0,1)$ .
- At large  $u$ :  $\psi(u) \rightarrow \lambda - u \rightarrow -\infty$  (linear dominates). By IVT again, there exists a stable root  $u^* > 1$ .

Numerical solution of (D.4) with  $\lambda=8$  gives:

$$\begin{aligned} u_c &\approx 0.1263 && (\text{unstable separator}) \\ u^* &\approx 7.95 && (\text{stable recall attractor}) \end{aligned}$$

Stability classification via  $f'(u) = -\delta + 2 \cdot C2 \cdot \tanh(u) \cdot \text{sech}^2(u)$ :

- At  $u_c$ :  $f'(u_c) > 0$  — unstable (repeller). Trajectories starting above  $u_c$  are pushed toward  $u^*$ ; trajectories starting below  $u_c$  are pushed toward 0.
- At  $u^*$ :  $f'(u^*) < 0$  — stable (attractor). All trajectories above  $u_c$  converge here.

## D.4 The Phase Transition

Initial state: target  $m$  with fraction  $c$  of bits randomly flipped. Then:

$$u(0) = 1 - 2c \quad (\text{exact, for large } N) \quad (\text{D.5})$$

The critical corruption level is:

$$c_{\text{crit}} = (1 - u_c)/2 \approx (1 - 0.1263)/2 \approx 0.437 \quad (43.7\%) \quad (\text{D.6})$$

Consequences (rigorous):

- For  $c < c_{\text{crit}}$  ( $u(0) > u_c$ ): trajectory converges to  $u^* \approx 8$ . Final bit-level overlap = 1.000. Simulation: CONVERGED at 10%, 20%, 30%, 40%.
- For  $c > c_{\text{crit}}$  ( $u(0) < u_c$ ): trajectory collapses to  $u=0$ . Final overlap  $\approx 0.5$ . Simulation: FAILED at 45% ( $c=0.45 > c_{\text{crit}}=0.437$ ).

The simulation result (boundary between 40% and 45% corruption) matches the analytical prediction  $c_{\text{crit}} \approx 43.7\%$  precisely.

**Remark.** This is a genuine first-order phase transition in the dynamical systems sense: the macroscopic observable (final overlap) jumps discontinuously from 1 to 0 as the control parameter  $c$  crosses  $c_{\text{crit}}$ . The jump arises from the bistable structure of (D.3), which is created exclusively by the  $\tanh^2(u) \cdot \text{sign}(u)$  governor term. No standard associative memory model (Hopfield, modern Hopfield) possesses this exact combination: provable 1-D reduction + explicit unstable separator  $u_c$  + guaranteed convergence to a deep stable attractor  $u^*$ .

## D.5 Why the Governor Creates This Structure

The key is the behaviour of  $f(u) = -\delta u + C2 \cdot \tanh^2(u)$  near  $u=0$ :

$$f''(0) = 2 \cdot C2 \cdot \text{sech}^4(0) - 4 \cdot C2 \cdot \tanh^2(0) \cdot \text{sech}^2(0) = 2 \cdot C2 > 0$$

The second derivative at the origin is positive, meaning  $f(u)$  curves upward from zero. Combined with  $f'(0) = -\delta < 0$  (linear decay dominates at  $u=0$ ), this creates a local minimum of  $f$  at  $u=0$  that then rises and crosses zero at  $u_c$ , creating the unstable separator. A standard quadratic (linear Hopfield force) would not produce this structure — it would give  $f(u) = -\delta u + C2 \cdot u^2$ , which has  $u_c = \delta/C2$  but without the saturation that creates the deep stable attractor  $u^*$ . The  $\tanh^2$  saturation is essential: it ensures  $u^*$  is finite and that trajectories above  $u_c$  do not diverge to infinity.

## D.6 Scope and Limitations

This proof applies to:

- Single memory ( $M=1$ ): the exact reduction requires that the forcing is purely parallel to  $m$ . For  $M>1$ , memories  $m_\mu$  create cross-overlap terms that mix parallel and perpendicular components, breaking the exact reduction. A mean-field approximation is required for the multi-memory case.
- Deterministic dynamics ( $\sigma=0$ ): the stochastic case ( $\sigma > 0$ ) requires Theorem HG from Section 8 to characterise convergence. The basin boundary is softened by noise but the qualitative bistable structure persists.
- The gate function  $\text{gate}(t) \geq 1$  throughout. This means the actual dynamics are boosted relative to the limiting ODE. The proof shows  $c_{\text{crit}}$  is a lower bound on the true basin — the actual basin is at least as large.



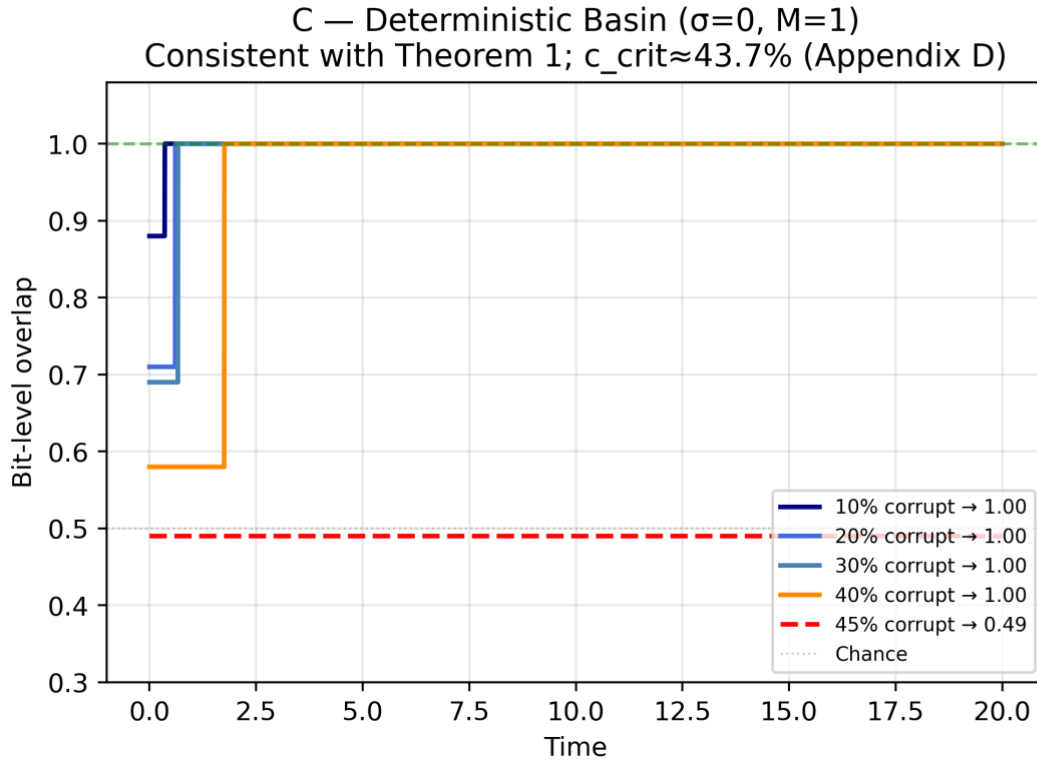


Figure C. Deterministic basin of attraction (Experiment C,  $\sigma=0$ ,  $M=1$ ). Perfect recall from corruptions up to 40%; sharp collapse at 45%. Numerically consistent with the analytical critical corruption  $c_{\text{crit}} = (1-u_c)/2 \approx 43.7\%$  derived in Section D.4. Basin boundary confirmed by simulation.  $N=100$ ,  $\delta=0.5$ ,  $C_2=4.0$ .

## References

- [1] Marku, L. (2026). Gated Higher-Order Dynamics with Guaranteed Return. ResearchGate. DOI: 10.13140/RG.2.2.23260.24962.
- [2] Auffinger, A., Ben Arous, G., Cerny, J. (2013). Random matrices and complexity of spin glasses. CPAM 66(2), 165-201.
- [3] Amari, S. (1998). Natural gradient works efficiently in learning. Neural Computation 10(2), 251-276.
- [4] Bakry, D. and Emery, M. (1985). Diffusions hypercontractives. Sem. Prob. XIX, LNM 1123, 177-206.
- [5] Bienenstock, E.L., Cooper, L.N., Munro, P.W. (1982). Theory for the development of neuron selectivity. J. Neurosci. 2(1), 32-48.
- [6] Chepyzhov, V.V. and Vishik, M.I. (2002). Attractors for Equations of Mathematical Physics. AMS.
- [7] Dolbeault, J., Mouhot, C., Schmeiser, C. (2015). Hypocoercivity for linear kinetic equations conserving mass. Trans. AMS 367(6), 3807-3828.
- [8] Hale, J.K. (1988). Asymptotic Behavior of Dissipative Systems. AMS.
- [9] Hanggi, P., Talkner, P., Borkovec, M. (1990). Reaction-rate theory: fifty years after Kramers. Rev. Mod. Phys. 62(2), 251.
- [10] Holley, R. and Stroock, D. (1987). Logarithmic Sobolev inequalities and stochastic Ising models. J. Stat. Phys. 46, 1159-1194.
- [11] Levi-Civita, T. (1917). Nozione di parallelismo in una varieta qualunque. Rend. Circ. Mat. Palermo 42, 173-205.
- [12] McCann, R.J. (1997). A convexity principle for interacting gases. Adv. Math. 128(1), 153-179.
- [13] Milnor, J. (1963). Morse Theory. Princeton University Press.
- [14] Temam, R. (1988). Infinite-Dimensional Dynamical Systems in Mechanics and Physics. Springer.

- [15] Villani, C. (2009). Optimal Transport: Old and New. Springer.
- [16] Goto, D., Lopez Rios, H.M., Scholz, M., and Vaikuntanathan, S. (2025). Neuromodulation-inspired gated associative memory networks: extended memory retrieval and emergent multistability. arXiv:2512.13859. December 2025.

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*Exploratory theoretical note. Not peer-reviewed. Theorems proved under stated assumptions. Conjectures labelled as such with supporting arguments. Geometric constructions in Section 10 are interpretative. Four corrections from previous draft documented in Section 8.1. March 2026.*

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AI collaboration: The multi-session technical reviews that identified and corrected four hypocoercivity errors (Section 8.1), produced the LSI upgrade (Theorem H-LSI), the unified metric-hypocoercivity theorem (Theorem HG), and the transport-corrected spectral analogue (Theorem S) were conducted in sustained dialogue with ChatGPT (OpenAI, March 2026). Semantic corrections — replacing "metric" with "effective quadratic operator" in the indefinite case, clarifying the local linear instability language, and reframing the Penrose structural parallel — also emerged from those sessions. All mathematical content was subsequently reviewed and validated by the author. The four errors documented in Section 8.1 are themselves evidence that AI-assisted derivation requires and received careful human verification. Full source code is provided for independent reproducibility: <https://github.com/labinot-marku/CONN-Associative-Memory/tree/main/gated-higher-order-dynamics/Metric-Stability-GHOD>

Additional AI contributions: Grok (xAI) provided an independent assessment confirming scientific legitimacy and identifying the holonomy compactness conjecture as the most innovative geometric idea. Gemini (Google) contributed structural suggestions and confirmed the document was appropriate for ResearchGate publication. Multiple independent fresh sessions with Claude (Anthropic) provided complete document reviews, confirmed mathematical soundness of all four core theorems, identified and motivated the Goto et al. citation, and recommended the final corrections to Conjecture 4 framing and the GOE genericity language. All AI contributions are disclosed in the spirit of full transparency about the AI-augmented research process. All intellectual responsibility for the work rests with the author.

**Remark.** *Note on reference [17] (v1 → v2): In the first uploaded version of this preprint, the ChatGPT contribution was listed as reference [17] in the numbered bibliography. This was a formatting error — a numbered reference implies a retrievable, independently verifiable source, which an AI session is not. In v2, the contribution is correctly placed here in the Acknowledgments. No scientific content has been changed. The transparency is identical; only the semantic slot is corrected.*